Fundamental Group of the Circle

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of fundamental group of the circle, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

AT-01 / Algebraic Topology / Report v1

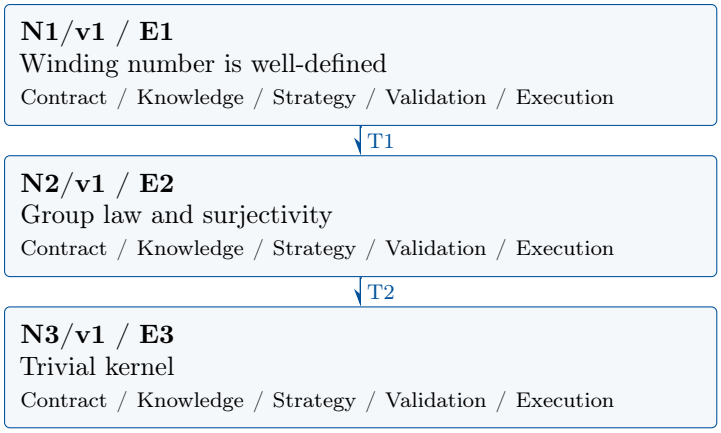
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Prove* $\pi_1(S^1, 1) \cong \mathbb{Z}$ *using the covering map* $p : \mathbb{R} \to S^1$, $p(t) = e^{2\pi it}$.

A. Foundations and permitted imports

Use the covering $p : R \to S^1$, $p(t) = e^{(2\pi it)}$. Import path lifting with prescribed initial value and homotopy lifting for a covering map: a path, or a homotopy of paths, has a unique continuous lift after an initial lift is fixed. Here evenly covered arcs supply the local inverses, and compact subdivision supplies global lifting. Based loops start/end at 1; their chosen lifts start at 0.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. An integer winding invariant on $\pi_1(S^1, 1)$.

Receiving activity. N2: Group law and surjectivity

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Two lifted loops and their endpoints; prior exports E1/v1

T2: N2/E2/v1 to N3. A surjective homomorphism $w : \pi_1(S^1, 1) \to Z$.

Receiving activity. N3: Trivial kernel

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: A lift with endpoint0; prior exports E2/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Winding number is well-defined

Object and role. Read the integer endpoint and audit based homotopies.

Inputs. A based loop α and its lift $\widetilde{\alpha}$; foundational imports stated in §1

C — Contract

Own E1/v1: An integer winding invariant on $\pi_1(S^1, 1)$.

K — Knowledge

Retain A based loop α and its lift $\widetilde{\alpha}$; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Read the integer endpoint and audit based homotopies.

V — Validation

The basepoint must remain fixed during the homotopy.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Since $p(\widetilde{\alpha}(1)) = 1$, the endpoint lies in Z.

N1.2 For a based homotopy, its lift has initial endpoint0 throughout; the terminal lifted endpoint is a continuous integer-valued function of the homotopy parameter, hence constant.

N1.3 Thus $w([\alpha]) = \widetilde{\alpha}(1)$ depends only on the based homotopy class.

Export E1/v1. An integer winding invariant on $\pi_1(S^1, 1)$.

Acceptance gate. The basepoint must remain fixed during the homotopy. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: The basepoint must remain fixed during the homotopy. Recheck dependent nodes N2, N3.

B. N2/v1: Group law and surjectivity

Object and role. Translate the second lift before concatenating.

Inputs. Two lifted loops and their endpoints; prior exports E1/v1

C — Contract

Own E2/v1: A surjective homomorphism $w : \pi_1(S^1, 1) \to Z$.

K — Knowledge

Retain Two lifted loops and their endpoints; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Translate the second lift before concatenating.

V — Validation

Concatenating two lifts both based at 0 would create a discontinuity unless the first winding is 0.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 If lifts of α and β end at m andn, a lift of the concatenation is the lift of α followed by the lift of β translated bym; its endpoint is m+n.

N2.2 Hencew is a homomorphism.

N2.3 For each $n \in Z$ the loop $\alpha_n(t) = e^{(2\pi i n t)}$ has liftnt and windingn, proving surjectivity.

Export E2/v1. A surjective homomorphism $w : \pi_1(S^1, 1) \to Z$.

Acceptance gate. Concatenating two lifts both based at 0 would create a discontinuity unless the first winding is 0. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Concatenating two lifts both based at 0 would create a discontinuity unless the first winding is 0. Recheck dependent nodes N3.

C. N3/v1: Trivial kernel

Object and role. Contract the closed lifted loop and project.

Inputs. A lift with endpoint0; prior exports E2/v1

C — Contract

Own E3/v1: The fundamental group of the circle with its explicit winding isomorphism.

K — Knowledge

Retain A lift with endpoint0; prior exports E2/v1; keep their assumptions and versions attached.

S — Strategy

Contract the closed lifted loop and project.

V — Validation

A projected contraction must preserve both loop endpoints.

E — Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 If $w([\alpha]) = 0$ then $\widetilde{\alpha}(0) = \widetilde{\alpha}(1) = 0$.

N3.2 The homotopy $H(s,t) = p((1-s)\widetilde{\alpha}(t))$ is continuous, remains based at 1 at both endpoints, equals α when $s = 0$, and is the constant loop when $s = 1$.

N3.3 Thus α is null-homotopic.

N3.4 The kernel is trivial, so the surjective homomorphism is injective and $\pi_1(S^1, 1) \cong Z$.

Export E3/v1. The fundamental group of the circle with its explicit winding isomorphism.

Acceptance gate. A projected contraction must preserve both loop endpoints. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: A projected contraction must preserve both loop endpoints. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Winding number is well-defined

For a based loop α, let $\widetilde{\alpha}$ be its unique lift starting at zero.

1. Since $p(\widetilde{\alpha}(1)) = 1$, the endpoint lies in Z.
2. For a based homotopy, its lift has initial endpoint0 throughout; the terminal lifted endpoint is a continuous integer-valued function of the homotopy parameter, hence constant.
3. Thus $w([\alpha]) = \widetilde{\alpha}(1)$ depends only on the based homotopy class.

B. §4.N2 — Group law and surjectivity

1. If lifts of α and β end at m andn, a lift of the concatenation is the lift of α followed by the lift of β translated bym; its endpoint is m+n.
2. Hencew is a homomorphism.
3. For each $n \in Z$ the loop $\alpha_n(t) = e^{(2\pi int)}$ has liftnt and windingn, proving surjectivity.

C. §4.N3 — Trivial kernel

1. If $w([\alpha]) = 0$ then $\widetilde{\alpha}(0) = \widetilde{\alpha}(1) = 0$.
2. The homotopy $H(s,t) = p((1-s)\widetilde{\alpha}(t))$ is continuous, remains based at 1 at both endpoints, equals α when $s = 0$, and is the constant loop when $s = 1$.
3. Thus α is null-homotopic.
4. The kernel is trivial, so the surjective homomorphism is injective and $\pi_1(S^1, 1) \cong Z$.

□

V. WORKED ORCHESTRATION

The loop $e^{(6\pi it)}$ has lift3t and winding3. Reversing it gives -3; concatenating it with $e^{(-4\pi it)}$ gives winding1. The covering itself identifies endpoint changes rather than relying on a picture of turns.

A. Boundary and hypothesis audit

An open path ending at -1 lifts to an endpoint in 1/2+Z, notZ, so it does not define this loop invariant. Validation checks the endpoint and based-homotopy conditions before exporting an integer.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

An open path ending at -1 lifts to an endpoint in 1/2+Z, notZ, so it does not define this loop invariant. Validation checks the endpoint and based-homotopy conditions before exporting an integer.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading.

- Allen Hatcher, Algebraic Topology: covering spaces, cellular homology, and cap products.